Printed Pages:03 **Sub Code: RME 075**

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B. TECH
(SEM VII) THEORY EXAMINATION 2022-23
OPERATIONS RESEARCH

Time: 3 Hours ***Total Marks: 70***

Note: Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt *all* questions in brief. **2x7 = 14**

(a) Define OR and discuss its scope.
(b) Differentiate between feasible solution and optimal solution.
(c) What is meant by an optimality test in a transportation problem?
(d) What is rule of Dominance?
(e) Why inventory is maintained?
(f) Mention the types of service disciplines.
(g) What are dummy activities in any Network technique?

SECTION B

2. Attempt any *three* of the following: **7x3 = 21**

(a) Two kinds of food F_1 and F_2 for animals are to be purchased. Food F_1 costs Rs. 20 a unit while food F_2 is available at Rs. 40 per unit. The nutrient contents per unit of these foods are as follows:

Nutrients	Nutrient Content	
	Food F_1	Food F_2
A	4	1
B	1	4
C	6	1

The minimum requirement of three nutrients is respectively, 20, 12 and 18 units. Formulate the problem as LPP and find out graphically the quantities of food units which should be bought at minimum cost.

(b) Discuss the steps of Hungarian method

(c) There are nine jobs, each of which must be processed on two machines I and II in the order I II. Processing time in hours are given below:

	A	B	C	D	E	F	G	H	I
M/c I	4	7	6	11	8	10	9	7	6
M/c II	8	10	9	6	5	11	5	10	13

Determine the sequence for the jobs that will minimize the total elapsed time T.

(d) A bakery keeps stock of popular brands of cake. Previous experiences show the daily demand pattern for the item associated probabilities, as given below:

Daily demand (number)	0	10	20	30	40	50
Probability	0.01	0.2	0.15	0.50	0.12	0.02

Use the following sequence of random numbers to simulate the demand for next 10 days. Random numbers: 25, 39, 65, 76, 12, 05, 73, 89, 19, 49
Also estimated the daily average demand for the cakes on the basis of simulated data.

(e) What is PERT? Define optimistic time, pessimistic time and most likely time and explain how you will estimate the expected time to complete the activity in PERT technique.

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SECTION C

3. Attempt any *one* part of the following: 7x1 = 7

(a) Use simplex method to solve

Minimize $Z = 3x + 8y$

Subject to $x + y = 200$

$x \leq 80$

$y \geq 60$

$x, y \geq 0$

(b) Obtain the dual of

Maximize $Z = 3X_1 - 2X_2$

$X_1 + X_2 \leq 5$

$X_1 \leq 4$

$X_2 \leq 6$

$X_2 \geq 1$

$X_1, X_2 \geq 0$

4. Attempt any *one* part of the following: 7x1 = 7

(a) A company has three plants A, B and C and three warehouse P, Q and R. The transportation cost per unit, demand of each warehouse and capacity of each warehouse and capacity of each plant are as given in the table below. Find the optimum transportation plan.

Plant	Warehouses			Capacity
	P	Q	R	
	Transportation cost (Rs.)			
A	50	80	100	400
B	22	90	40	500
C	70	100	55	300
Demand	400	400	400	1200

(b) Solve the following minimal assignment problem

Job	Man			
	A	B	C	D
I	12	30	21	15
II	18	33	9	31
III	44	25	24	21
IV	23	30	28	14

5. Attempt any *one* part of the following: 7x1 = 7

(a) A and B plays a game in which each has three coins: 5 paise, 10 paise and 20 paise. Each selects a coin without the knowledge of other's choice. If the sum of the coin is an odd amount, A wins B's coin and if the sum is even, B wins A's coin. Find the best strategy for each player and the value for the game.

(b) Give Johnson's procedure for determining an optimal sequence for processing n items on two machines. Give justification of the rule used in the procedure.

6. Attempt any *one* part of the following: 7x1 = 7

(a) A factory needs 36000 units annually of a component that cost Rs. 2 per unit. Cost of each order placing is Rs. 25 and inventory carrying cost is Rs. 10 per year.

(i) Find the economic lot size and the total inventory cost.

(ii) What is the time between placing of orders?

(iii) The supplier offers 2% discount if a single order is placed.

(b) Write short note on Simulation.

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7. Attempt any *one* part of the following: 7x1 = 7

(a) An automatic vending machine at the airport dispenses a cup of coffee in 48 seconds. Customers arrive according to Poisson process with a mean rate of 25 per hour. Calculate (i) the average number of people that will be waiting for service (ii) The average waiting time of the customer (iii) Average number of customers in the system.

(b) A small scale industrial unit consists of 6 activities as given below:

Activity	Time in days	Pre-operation
A	5	None
B	6	A
C	5	B
D	4	A
E	3	D
F	4	C,E

Draw the network diagram and calculate EST, LST, EFT, LFT and floats, mark the critical path and find total project duration.

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